Robogarage Digital Twin with Isaac Sim + ROS2

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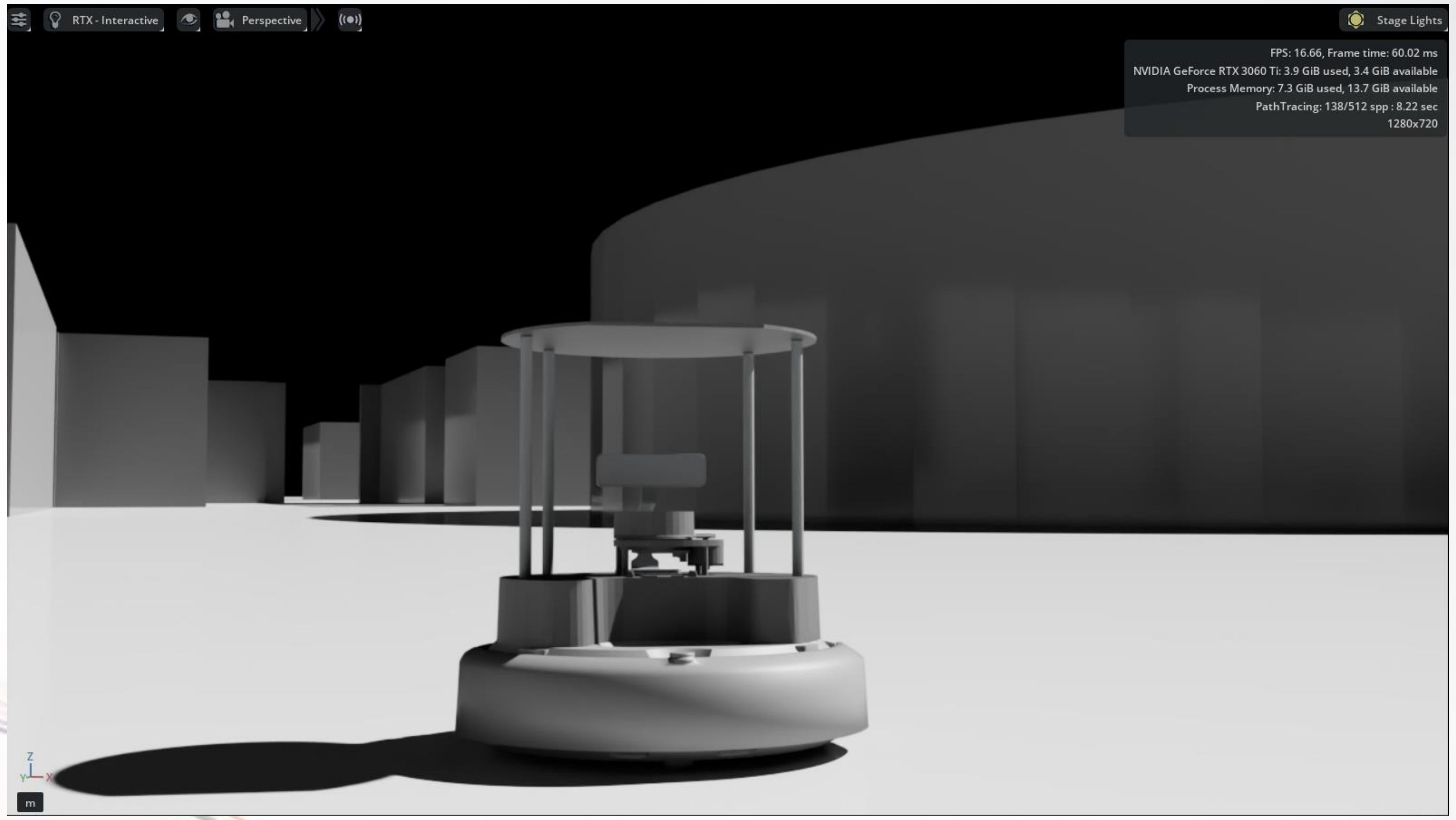
This project creates a simulated environment, allowing a robot to move, navigate, sensor-scan and visualize the digital twin space before real-world deployment.



Description:

This project builds a virtual campus environment where a robot can move, navigate, and scan its surroundings. By combining a campus map, robot simulation, ROS 2 communication, and sensor visualization, the project creates a safe testing platform for future campus robot applications.

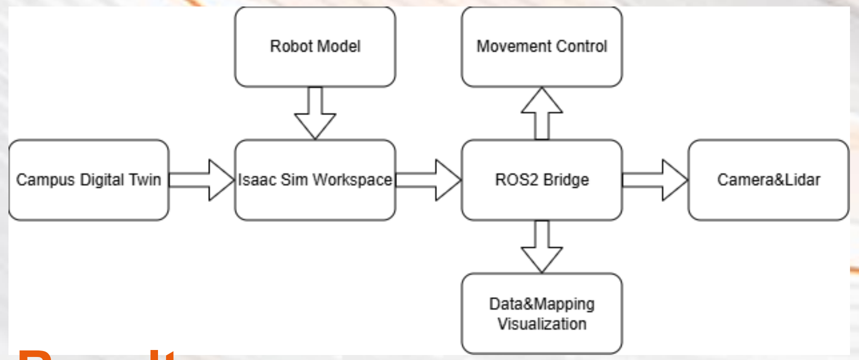
Objective:

- Put a robot into a simulated digital twin place.
- Implement robot control and navigation.
- Add simulated Camera and Sensors to the robot.
- Real time data visualization.
- Generate and save scanning results.

Technical Challenges:

- A campus-based virtual simulation environment
- A robot model configured inside the map
- A movement test setup for route-based navigation
- A simulated LiDAR scanning function
- SLAM Map generation
- Foxglove data visilazation

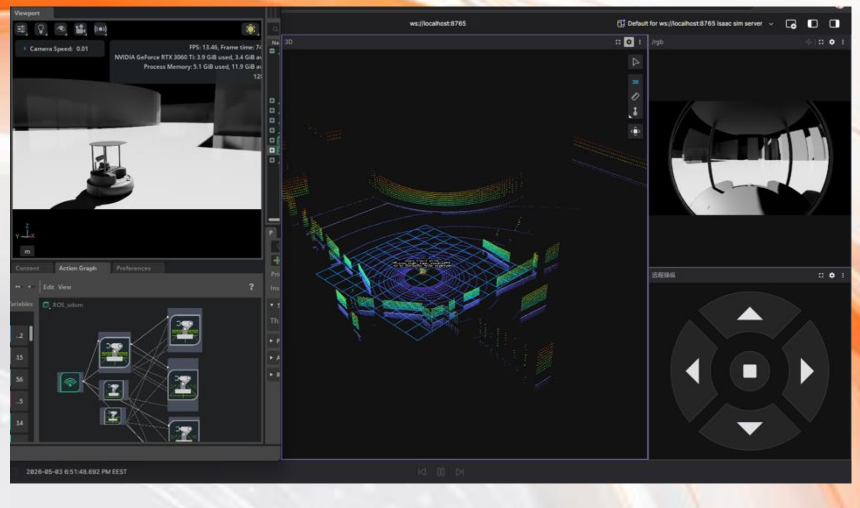
Technical Pipeline:



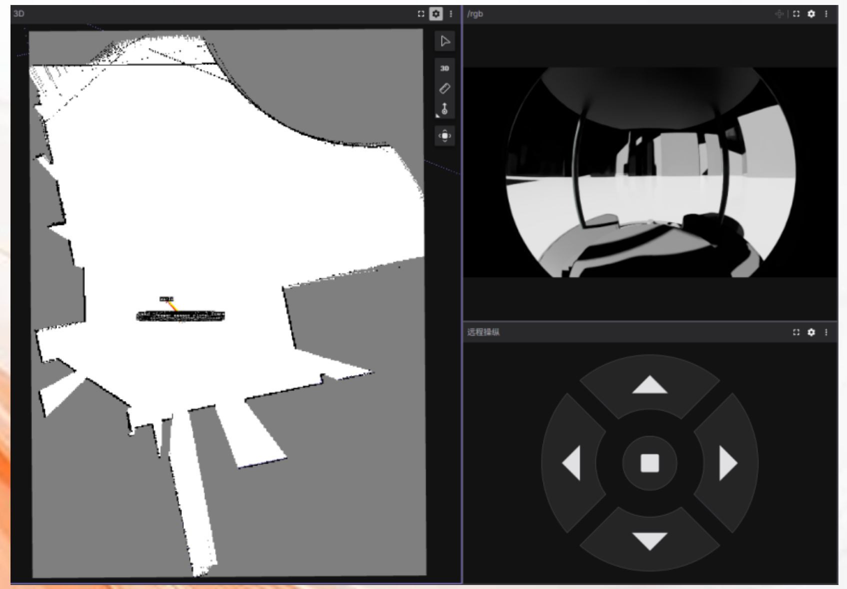
Results:

- Robot placed successfully in the virtual campus inside Isaac Sim.
- Basic movement control using ROS2 bridge.
- Simulated Camera and 2D & 3D LiDAR scan.
- Data visualized through ROS2 & Foxglove Bridge.
- Real time virtual campus environment mapping using SLAM.
- Prototype prepared for live demonstration.

TurtleBot4 Introduction



Foxglove data visualization interface



RViz / LiDAR scan screenshot

Demo video

GitHub repo

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